ROAD TRAFFIC ACCIDENT ANALYSIS

Data Dictionary & DAX Measures

Graduation Project - DEPI Data Analysis Track | Team 4

1. Data Dictionary

The dictionary describes every column of the dataset: its meaning and data type. Unknown marks a value that was not recorded.

Column	Data type	Meaning
Time	Time	Time of day of the accident (hh:mm:ss).
Day_of_week	Text	Day of the week of the accident.
Age_band_of_driver	Text	Age group of the driver.
Sex_of_driver	Text	Sex of the driver.
Educational_level	Text	Education level of the driver.
Vehicle_driver_relation	Text	Relation of the driver to the vehicle.
Driving_experience	Text	Driving experience of the driver, in year bands.
Type_of_vehicle	Text	Type of the vehicle involved in the accident.
Owner_of_vehicle	Text	Type of the vehicle owner.
Service_year_of_vehicle	Text	Years the vehicle has been in service, in year bands.
Defect_of_vehicle	Text	Whether the vehicle had a defect: No defect, or a defect code.
Area_accident_occured	Text	Type of area where the accident happened.
Lanes_or_Medians	Text	Lane or median layout of the road.
Road_allignment	Text	Shape and terrain of the road.
Types_of_Junction	Text	Type of junction at the accident location.
Road_surface_type	Text	Material of the road surface.
Road_surface_conditions	Text	Condition of the road surface at the time.
Light_conditions	Text	Light at the time of the accident.
Weather_conditions	Text	Weather at the time of the accident.
Type_of_collision	Text	How the collision happened.
Number_of_vehicles_involved	Whole number	Number of vehicles in the accident.
Number_of_casualties	Whole number	Number of casualties in the accident.
Vehicle_movement	Text	What the vehicle was doing before the accident.
Casualty_class	Text	Class of the casualty. Unknown means no casualty details were recorded.
Sex_of_casualty	Text	Sex of the casualty.
Age_band_of_casualty	Text	Age group of the casualty.
Casualty_severity	Text	Severity code of the casualty.
Work_of_casuality	Text	Work of the casualty.
Fitness_of_casuality	Text	Physical fitness of the casualty.
Pedestrian_movement	Text	Movement of the pedestrian, or Not a Pedestrian.
Cause_of_accident	Text	Main cause of the accident (20 categories).
Accident_severity	Text	Severity of the accident: the main outcome analysed in the project.
Hour (derived from Time)	Whole number	Hour of the day (0 to 23) taken from Time. Used for the hourly analysis.

2. DAX Measures

The table lists each measure with its purpose, the business questions it answers, the columns and DAX functions it uses, its benefit, and the dashboard page where it appears.

#	Measure name	Purpose	Business question No.	Columns used	DAX functions	Benefit	Dashboard page
1	**Total Accidents**	Counts the accidents in the current filter selection.	All (base); KPI	Accident rows	COUNTROWS	Base count used by all other measures.	All pages
2	**Serious or Fatal Accidents**	Counts accidents with Serious or Fatal severity.	2, 5, 10, 24, 27, 29, 34	Accident_severity	CALCULATE, IN	Numerator of the main severity percentage.	Overview, scope pages
3	**Serious or Fatal %**	Percentage of accidents that are serious or fatal.	2, 5, 24, 27, 29; KPI	Accident_severity	DIVIDE	Compares groups of different sizes fairly.	Overview, scope pages
4	**Fatal Accidents**	Counts accidents with Fatal injury.	40; KPI	Accident_severity	CALCULATE	Shows the most severe outcome on its own.	Overview
5	**Serious Accidents**	Counts accidents with Serious Injury.	40	Accident_severity	CALCULATE	Completes the severity split.	Overview
6	**Slight Accidents**	Counts accidents with Slight Injury.	40	Accident_severity	CALCULATE	Completes the severity split.	Overview
7	**% of All Accidents**	Share of each group out of all accidents in the current selection.	8, 35, 40	Column on the visual axis	DIVIDE, CALCULATE, ALLSELECTED	One measure gives the share for any category and follows the slicers.	Overview, Driver, Causes and Casualties
8	**Serious or Fatal % (30+ Accidents)**	Serious or Fatal % shown only for groups with at least 30 accidents.	10, 34	Accident_severity	IF	Keeps very small groups out of the rankings.	Vehicle, Causes and Casualties
9	**No Licence Accidents**	Counts accidents where the driver has no licence.	7	Driving_experience	CALCULATE	Gives a direct number for a card.	Driver
10	**Accidents with Recorded Defect**	Counts accidents where a vehicle defect is recorded.	13	Defect_of_vehicle	CALCULATE, NOT, IN	Shows how many accidents involve a vehicle defect.	Vehicle
11	**Average Vehicles per Accident**	Average number of vehicles in an accident.	15; KPI	Number_of_vehicles_involved	AVERAGE	Shows how complex a typical accident is.	Overview, Vehicle
12	**Multi-Vehicle Accidents %**	Percentage of accidents with more than two vehicles.	16	Number_of_vehicles_involved	DIVIDE, CALCULATE	Separates simple accidents from large ones.	Vehicle

#	Measure name	Purpose	Business question No.	Columns used	DAX functions	Benefit	Dashboard page
13	**Junction Accidents %**	Percentage of accidents at a junction, among accidents with a known junction type.	18	Types_of_Junction	DIVIDE, CALCULATE, NOT, IN	Shows how much of the problem is linked to junctions.	Road and Junction
14	**Peak Hour**	The hour of the day with the most accidents in the current selection.	25; KPI	Time (hour of the day)	MAXX, TOPN, VALUES	One clear number for the KPI card.	Overview, Environment and Time
15	**Daylight %**	Percentage of accidents that happen in daylight.	28	Light_conditions	DIVIDE, CALCULATE	Summarises light conditions in one number.	Environment and Time
16	**Darkness %**	Percentage of accidents that happen in darkness.	28	Light_conditions	DIVIDE, CALCULATE	Pairs with Daylight % (together 100%).	Environment and Time
17	**Weekend Accidents %**	Percentage of accidents on Saturday and Sunday.	32	Day_of_week	DIVIDE, CALCULATE, IN	Compares weekend and weekday risk.	Environment and Time
18	**Top Cause**	The most frequent cause of accident in the current selection.	33; KPI	Cause_of_accident	MAXX, TOPN, VALUES	One clear label for the KPI card that follows the slicers.	Overview, Causes and Casualties
19	**Cause Rank**	Rank of each cause by number of accidents.	33, 35	Cause_of_accident	RANKX, ALLSELECTED	Shows the position of each cause next to its count.	Causes and Casualties
20	**Average Casualties per Accident**	Average number of casualties in an accident.	36; KPI	Number_of_casualties	AVERAGE	Shows how many people a typical accident affects.	Overview, Causes and Casualties
21	**Multi-Casualty Accidents %**	Percentage of accidents with more than one casualty.	37	Number_of_casualties	DIVIDE, CALCULATE	Shows how often an accident hurts more than one person.	Causes and Casualties
22	**Casualty Records**	Counts accidents that have casualty details.	38, 39	Casualty_class	CALCULATE	Casualty questions use only accidents that have casualty details.	Causes and Casualties
23	**Casualty Share %**	Share of each casualty group out of all casualty records.	38, 39	Casualty_class, Age_band_of_casualty	DIVIDE, CALCULATE, ALLSELECTED	Works for any casualty column on the axis.	Causes and Casualties

3. Business Questions and the Measures That Answer Them

Each of the 40 business questions is answered by one or two of the measures in section 2.

A. Driver (Driver page)

No.	Business question	Measure(s) used
1	Which driver age band is involved in the most accidents?	Total Accidents
2	Which driver age band has the highest percentage of serious or fatal accidents?	Serious or Fatal %
3	How many accidents involve male drivers and how many involve female drivers?	Total Accidents
4	Which driving experience group has the most accidents?	Total Accidents (+ Serious or Fatal %)
5	Which driving experience group has the highest percentage of serious or fatal accidents?	Serious or Fatal %
6	Which education level has the most accidents?	Total Accidents
7	How many accidents involve drivers with no licence?	No Licence Accidents
8	What share of accidents comes from each driver-vehicle relation?	% of All Accidents

B. Vehicle (Vehicle page)

No.	Business question	Measure(s) used
9	Which vehicle type is involved in the most accidents?	Total Accidents
10	Which vehicle type has the highest percentage of serious or fatal accidents (30+ accidents)?	Serious or Fatal % (30+ Accidents)
11	Which vehicle owner type has the most accidents?	Total Accidents
12	Which vehicle service-year group has the most accidents?	Total Accidents
13	How many accidents involve a vehicle with a recorded defect?	Accidents with Recorded Defect
14	Which vehicle movement is the most common in accidents?	Total Accidents
15	What is the average number of vehicles involved per accident?	Average Vehicles per Accident
16	What percentage of accidents involve more than two vehicles?	Multi-Vehicle Accidents %

C. Road and Junction (Road and Junction page)

No.	Business question	Measure(s) used
17	Which junction type has the most accidents?	Total Accidents
18	What percentage of accidents happens at a junction?	Junction Accidents %
19	Which lane or median type has the most accidents?	Total Accidents
20	Which road alignment has the most accidents?	Total Accidents
21	Which road surface type has the most accidents?	Total Accidents
22	Which area type has the most accidents?	Total Accidents
23	Which type of collision is the most common?	Total Accidents
24	Which type of collision has the highest percentage of serious or fatal accidents?	Serious or Fatal %

D. Environment and Time (Environment and Time page)

No.	Business question	Measure(s) used
25	At which hour of the day do the most accidents happen?	Total Accidents + Peak Hour

No.	Business question	Measure(s) used
26	Which day of the week has the most accidents?	Total Accidents
27	Which day of the week has the highest percentage of serious or fatal accidents?	Serious or Fatal %
28	What percentage of accidents happens in daylight and what percentage in darkness?	Daylight %, Darkness %
29	Which light condition has the highest percentage of serious or fatal accidents?	Serious or Fatal %
30	Which weather condition has the most accidents?	Total Accidents
31	How many accidents happen on each road surface condition?	Total Accidents
32	What percentage of accidents happens on weekends (Saturday and Sunday)?	Weekend Accidents %

E. Causes and Casualties (Causes and Casualties page; Q40 on Overview)

No.	Business question	Measure(s) used
33	Which cause of accident is the most frequent?	Total Accidents + Top Cause + Cause Rank
34	Which five causes have the highest percentage of serious or fatal accidents (30+ accidents)?	Serious or Fatal % (30+ Accidents)
35	What share of all accidents does each of the top five causes represent?	% of All Accidents
36	What is the average number of casualties per accident?	Average Casualties per Accident
37	What percentage of accidents has more than one casualty?	Multi-Casualty Accidents %
38	What share of casualties are drivers or riders, passengers and pedestrians?	Casualty Share %
39	Which casualty age band has the most casualties?	Casualty Records (+ Casualty Share %)
40	How many accidents are Slight, Serious and Fatal, and what is the percentage of each?	Slight, Serious, Fatal Accidents + % of All Accidents

KPI cards (Overview page)

KPI card (Overview page)	DAX measure	Business question
Total Accidents	Total Accidents	Base of all questions
Serious or Fatal Share	Serious or Fatal %	2, 5, 10, 24, 27, 29, 34
Fatal Accidents	Fatal Accidents	40
Average Casualties	Average Casualties per Accident	36
Average Vehicles	Average Vehicles per Accident	15
Top Cause	Top Cause	33
Peak Hour	Peak Hour	25